Temporal Correlations Reshape Causal Resources in Qubit Hamiltonian Recognition

Exact Two-Query Strategy Comparison and a Uniform Critical Crossover

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Revised and extended preprint draft v1.2 — 16 October 2026
Not peer-reviewed; new Zenodo version DOI pending publication

Abstract

We investigate binary recognition of an unknown qubit Hamiltonian from two queries of its evolution, where the unknown evolution angles are either independently resampled or shared with tunable mixing strength η. For a symmetric nonorthogonal pair of Hamiltonians we compare parallel (PAR), sequential (SEQ), and general (GEN) quantum testers. A symmetry reduction identifies the full sequential optimum with optimization over a 3×3 correlation elliptope, and a legal nine-generator general-strategy dual provides an exact matching upper bound, $P_{\mathrm{GEN}} = P_{\mathrm{SEQ}}$. The parallel feasible set is a Bell tetrahedron within this elliptope. The different boundary geometries make arbitrarily weak temporal correlation operationally relevant, with a fixed-angle GEN–PAR gap proportional to η^2, $\eta^{4/3}$, or η below, at, or above the critical angle $\beta = \pi/4$. The *new extension relative to the author's archived v1.1 preprint* is a uniform double-scaling theorem in the bounded window $\beta = \pi/4 + \mu\eta^{2/3}$, with an explicit strictly positive crossover function. This is a result within standard quantum tester theory, not evidence for physics beyond quantum mechanics. The technical proofs and independently checked computations are supplied in the accompanying supplementary material.

Keywords: Hamiltonian recognition; nonorthogonal qubits; quantum testers; temporal correlations; sequential discrimination; semidefinite duality; critical asymptotics.

1 Introduction and relationship to prior work

Hamiltonian recognition distinguishes the generator of an unknown unitary evolution, while the evolution angle is not available to the observer. Zhu *et al.* introduced a general tester treatment and derived global optimality for orthogonal qubit generator axes, also numerically examining nonorthogonal axes [1]. Adaptive advantages in channel discrimination, including separation from parallel schemes, were established earlier [3, 4, 5]. Thus neither Hamiltonian recognition itself nor the general PAR–SEQ–GEN hierarchy is claimed here as new.

Our setting resolves a different, narrowly defined question: for *two* queries of a symmetric *nonorthogonal* Hamiltonian pair, how does the joint distribution of the inaccessible evolution times affect the exact value of causal processing? In particular, the model mixes the independent-angle and same-angle experiments. It is a conventional random-unitary two-slot quantum process. The author's previously archived v1.1 manuscript [2] treated the full two-query strategy phase diagram; later internal addenda established fixed-angle matching asymptotics. This revision contributes a bounded-window uniform critical crossover and a consolidated transparent proof audit. The new result is an extension of that preprint, not a second independent discovery of its existing theorems.

2 Physical model and tester hierarchy

Let X, Y, Z be Pauli matrices and choose

$$H_{\pm} = \pm \sin(\beta/2)X + \cos(\beta/2)Z, \quad U_H(\theta) = \cos\theta I - i \sin\theta H, \quad 0 \le \beta \le \pi/2. \tag{1}$$

The hypotheses have equal priors. An unknown θ is uniform on $[0, \pi]$. Two channel uses are sampled independently with probability $1-\eta$, or use the same sampled θ with probability $\eta \in [0, 1]$. Using unnormalized

vectorization, write $A = |I\rangle\!\rangle \otimes |I\rangle\!\rangle$, $B_1 = |I\rangle\!\rangle \otimes |H\rangle\!\rangle$, $B_2 = |H\rangle\!\rangle \otimes |I\rangle\!\rangle$, and $C = |H\rangle\!\rangle \otimes |H\rangle\!\rangle$. The averaged two-slot performance operators are

$$\Omega_H^{\mathrm{iid}} = \tfrac{1}{4}(AA^\dagger + B_1B_1^\dagger + B_2B_2^\dagger + CC^\dagger), \tag{2}$$

$$\Omega_H^{\mathrm{shared}} = \tfrac{3}{8}AA^\dagger + \tfrac{1}{8}(B_1+B_2)(B_1+B_2)^\dagger + \tfrac{3}{8}CC^\dagger - \tfrac{1}{8}(AC^\dagger + CA^\dagger), \tag{3}$$

$$\Omega_\pm(\eta) = (1-\eta)\Omega_{H_\pm}^{\mathrm{iid}} + \eta\Omega_{H_\pm}^{\mathrm{shared}}. \tag{4}$$

These are normalized performance operators for the physical random-unitary query model. Denote the optimal correct-guess probabilities by $P_{\mathrm{PAR}} \le P_{\mathrm{SEQ}} \le P_{\mathrm{GEN}}$, for parallel, fixed-order sequential and general two-slot testers.

3 Exact sequential and general-strategy optimization

Task symmetries and causal-tester normalization reduce the full sequential deterministic tester to a three-parameter family without loss [2]. The exact positive semidefiniteness condition is

$$G = \begin{pmatrix} 1 & x & z \\ x & 1 & -y \\ z & -y & 1 \end{pmatrix} \succeq 0. \tag{5}$$

Set $c = \cos\beta$, $s = \sin\beta$ and

$$\boldsymbol{w}_\eta = \big(-\eta, \tfrac{1}{2}(2+\eta)(1-c), \tfrac{1}{2}(2+\eta)(1+c)\big)^{\mathsf{T}}. \tag{6}$$

The full sequential objective, derived from the generalized Helstrom expression on 16×16 matrices, is

$$P_{\beta,\eta}(x,y,z) = \frac{1}{2} + \frac{s}{8}\left[\sqrt{(1+x)(1+z)} + (1-\eta)\sqrt{(1-x)(1-z)}\right] + \frac{s}{16}\sqrt{2(1+y)\,\boldsymbol{w}_\eta^{\mathsf{T}} G \boldsymbol{w}_\eta}. \tag{7}$$

Consequently,

$$P_{\mathrm{SEQ}}(\beta,\eta) = \max_{G\succeq 0} P_{\beta,\eta}(G). \tag{8}$$

The physical parallel subset is the Bell tetrahedron, parameterized by $p_j \ge 0$, $\sum_{j=0}^3 p_j = 1$ and

$$x = p_0 - p_1 + p_2 - p_3, \quad y = -p_0 + p_1 + p_2 - p_3, \quad z = p_0 + p_1 - p_2 - p_3. \tag{9}$$

Theorem 1 (Exact general-strategy closure; v1.1 result)**.** *For the processes in Eq. (1), throughout the stated parameter domain,*

$$P_{\mathrm{GEN}}(\beta,\eta) = P_{\mathrm{SEQ}}(\beta,\eta) = \max_{G\succeq 0} P_{\beta,\eta}(G). \tag{10}$$

Proof structure. After identifying the full general-strategy dual affine normalization, restrict to the legal nine-generator affine subspace of non-signalling two-slot dual certificates. In the normalized Pauli-vector basis, the support splits into sectors of dimensions $4, 2, 2, 1$, and the restricted dual is a nine-variable semidefinite program. Its Lagrange dual has a positive 3×3 Gram block G and three complementary positive blocks. Stationarity fixes their diagonal entries, while the allowed off-diagonal entries saturate 2×2 Schur-complement bounds. The optimal dual expression is precisely Eq. (7), including the fixed term $1/2$ and the vector square-root contribution. A strict feasible point implies strong SDP duality. The resulting legal general-strategy upper bound equals P_{SEQ}, while strategy inclusion gives the reverse bound. The precise nine Pauli generators, stationarity equations and normalization details are reproduced in the supplementary S8; a separate matrix-level audit is included in the reproducibility archive. □

4 Fixed-angle weak-correlation asymptotics

Introduce local coordinates

$$u = \frac{x+z}{2}, \qquad v = \frac{x-z}{2}, \qquad h = 1-y. \tag{11}$$

The elliptope determinant is exactly

$$\det G = 2h(1+u^2-v^2) - h^2 - 4u^2. \tag{12}$$

The tetrahedron imposes the stronger inequalities $h \pm 2u \ge 0$ and $2 - h \pm 2v \ge 0$. Thus locally the elliptope permits $h \asymp u^2$ while the tetrahedron enforces $h \ge 2|u|$.

Theorem 2 (Fixed-angle matching asymptotics; previously established in the author's September 2026 addendum)**.** *For fixed $0 < \beta < \pi/2$, define $\Delta = P_{\mathrm{GEN}} - P_{\mathrm{PAR}}$. As $\eta \downarrow 0$,*

$$\Delta(\beta, \eta) \sim \begin{cases} K_<(\beta)\eta^2, & 0 < \beta < \pi/4, \\ K_c\eta^{4/3}, & \beta = \pi/4, \\ K_>(\beta)\eta, & \pi/4 < \beta < \pi/2, \end{cases} \tag{13}$$

where, with $c = \cos\beta$, $s = \sin\beta$, $d = (2s^2 - 1)/(2s^2)$,

$$K_< = \frac{s(1-c)^2}{32c(2c^2-1)}, \quad K_c = \frac{3(2-\sqrt{2})}{64}\left(\frac{2-\sqrt{2}}{4}\right)^{1/3}, \quad K_> = \frac{1-s}{8}(\sqrt{d} - d). \tag{14}$$

These fixed-angle limits are not automatically uniform near $\beta = \pi/4$. That missing uniform statement is the main additional result developed below.

5 Exact phase boundaries and endpoint checks

The fixed-angle scaling is part of a larger exact two-parameter phase diagram developed in the earlier manuscript and Supplement S11–S14. In the low-angle regime the nontrivial interior equality curve is

$$\sin^2\beta = \frac{4\eta^2}{\eta^3 + \eta^2 + 4}, \qquad P_{\mathrm{null}}(\eta) = \frac{1}{2} + \frac{2\eta}{\eta^3 + \eta^2 + 4}. \tag{15}$$

This exceptional curve follows from Bell-edge stationarity combined with an outward tangent certificate for the complete elliptope. In the high-angle range $\beta_c \le \beta < \pi/2$ with $\beta_c = \arccos(1/\sqrt{3})$, $P_{\mathrm{GEN}} > P_{\mathrm{PAR}}$ for every $0 < \eta < 1$, yet equality is recovered at the two endpoints $\eta = 0, 1$: a reentrant positive-gap phase. On the fully shared-time edge $\eta = 1$, the strategy gap is strictly positive for $0 < \beta < \beta_c$ and vanishes for $\beta_c \le \beta \le \pi/2$. At the orthogonal endpoint $\beta = \pi/2$, all three strategy classes agree and

$$P_{\mathrm{GEN}} = P_{\mathrm{SEQ}} = P_{\mathrm{PAR}} = \frac{7 + 3\eta}{4(\eta + 2)}. \tag{16}$$

These global phase statements are distinct from the local crossover theorem below. The mathematical certificates, including the exceptional-edge and high-angle positivity arguments, reside in Supplement S11–S14 rather than being claimed as additional discoveries of the present revision.

6 Uniform double-scaling crossover: the added result

Write

$$t = \eta^{1/3}, \qquad \beta = \frac{\pi}{4} + \mu t^2, \qquad |\mu| \le M < \infty. \tag{17}$$

Define, with $a = 2 - \sqrt{2}$,

$$\boxed{\mathcal{F}(\mu) = \max_{q \ge 0} \frac{aq + 4\mu q^2 - q^4}{16} - \frac{(\max\{\mu, 0\})^2}{4}.} \tag{18}$$

Theorem 3 (Uniform critical crossover; v1.2 extension)**.** *For every finite M, uniformly over $|\mu| \le M$ as $\eta \downarrow 0$,*

$$\boxed{\Delta\left(\frac{\pi}{4} + \mu\eta^{2/3}, \eta\right) = \eta^{4/3}\mathcal{F}(\mu) + o_M(\eta^{4/3}).} \tag{19}$$

Moreover $\mathcal{F}(\mu) > 0$ for every finite real μ. The positive maximizer is the unique positive solution of

$$4q^3 - 8\mu q - a = 0. \tag{20}$$

Proof outline; the complete uniform estimates are in new Supplement S19. At $(\beta, \eta) = (\pi/4, 0)$, the exact objective separates into $A + B + Q$ with $A + B \le 2$, equality only for $v = 0$, and $Q(h) = \sqrt{2 - h^2/2}$, equality only for $h = 0$. The determinant condition then forces $u = 0$. Thus the critical global maximizer is unique. Compactness gives a positive objective gap outside any fixed neighborhood and hence places *all* small-t global optimizers in a common neighborhood uniformly over $|\mu| \le M$.

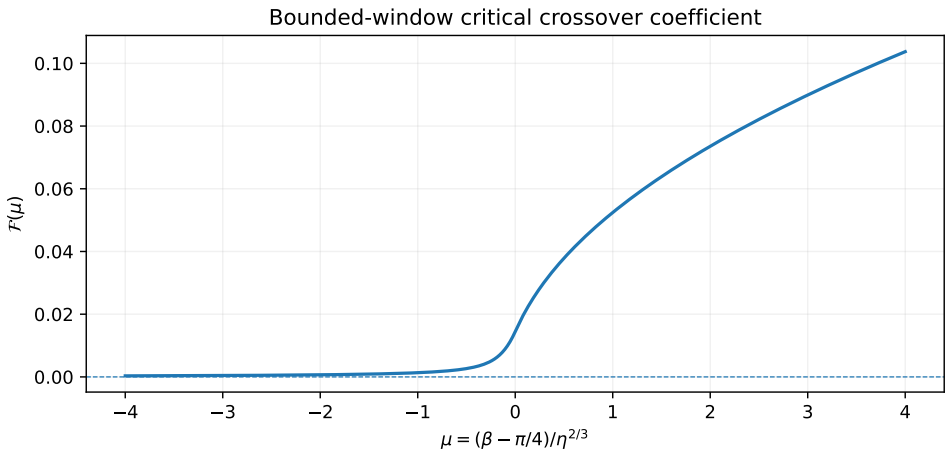


Figure 1: The crossover coefficient $\mathcal{F}(\mu)$ from Eq. (18). This analytic prediction is derived from the exact two-query optimization in the indicated shrinking window; it is not experimental data.

Inside this neighborhood, the objective is smooth. Its critical quadratic/quartic coercivity and the vanishing mixed derivatives $\partial_\beta \partial_u P = \partial_\beta \partial_v P = 0$ yield, at every global maximizer,

$$h^2 + v^2 + u^4 \le C_M\big(t^2 h + t^3|u| + t^3|v| + t^3 h\big). \tag{21}$$

Young inequalities give $u = O_M(t)$ and $v, h = O_M(t^2)$. Substitute $u = Ut$, $v = Vt^2$, $h = Ht^2$. The elliptope limit is $H \ge 2U^2$, and the expansion relative to the common feasible base point is

$$\frac{P_{\beta,\eta}(Ut, Vt^2, Ht^2) - P_{\beta,\eta}(0,0,0)}{t^4} \to \Psi_\mu(U,V,H) = -\frac{H^2}{64} + \frac{\mu H}{8} - \frac{aU}{16} - \frac{\sqrt{2}V^2}{16}, \tag{22}$$

uniformly on bounded blown-up sets and $|\mu| \le M$. The maximizing elliptope point has $V = 0$, $U = -\sqrt{H/2}$; putting $q = \sqrt{H/2}$ gives the first maximization in Eq. (18). This maximum is achieved by the *exactly feasible* path $u = -qt$, $v = 0$, $h = 2q^2t^2$, for which $\det G = 0$.

The Bell tetrahedron imposes $h \ge 2|u|$, so any bounded blown-up PAR optimizer has $U = 0$. Its optimal limit has $V = 0$, $H = 4\mu_+$ and value $\mu_+^2/4$, attained by an exactly feasible tetrahedral path. Uniform compact localization, bounded rescaled optimizers, uniform Taylor remainder, and these recovery paths yield the claimed two-sided uniform optimization asymptotics.

For positivity, if $\mu \le 0$ the positive term aq beats the higher-degree terms for sufficiently small $q > 0$. If $\mu > 0$, the positive maximizer obeys $r = q^2 - 2\mu = a/(4q) > 0$, and substitution gives $\mathcal{F}(\mu) = \mu r/2 + 3r^2/16 > 0$. □

Remark 1 (Scope of uniformity)**.** *Theorem 3 is uniform only for bounded μ; fixed-angle matching requires separate asymptotic limits. Taking $\mu \to \pm\infty$* after *the bounded-window theorem yields formal matching coefficients, not by itself a uniform overlap-region theorem.*

7 Interpretation, limitations and verifiability

The new noninteger critical exponents arise from a difference of feasible-set boundaries: near the critical point, the elliptope curvature scales as $h \sim u^2$ while the parallel tetrahedron enforces $h \gtrsim |u|$. Their responses to a small shared-time perturbation therefore differ. This is a standard-quantum resource statement, not an extra law of nature or a property implied by the author's separate speculative structural ontology.

The main originality candidate is the *particular exact two-query correlated-time nonorthogonal model*, its full strategy comparison and its double-scaling crossover. The most closely related published Hamiltonian-recognition work includes a theorem on orthogonal axes and numerical nonorthogonal calculations [1]. Global no-advantage and adaptive-advantage statements are well established in other channel-discrimination families [3, 5, 4]. The present result should not be advertised as a general nonorthogonal Hamiltonian-recognition theorem for arbitrary time laws, arbitrary query counts or general hidden-memory processes. Numerical checks reproduce selected exact identities but are not substitutes for analytic optimality.

Archival status. This is a prepared v1.2 revision of a pre-existing Zenodo preprint, awaiting author deposit and assignment of its own *version DOI*. The v1.1 record and date are not to be changed retrospectively. The proof verification and literature survey remain open to external review.

Acknowledgments and disclosure

The author is responsible for the scientific claims, proof statements, and final release decisions. ChatGPT was used as a drafting, symbolic-checking, and computational-audit assistant; its output is not independent human peer review.

Data and code availability

The accompanying archive includes (i) the full Supplement S1–S18 from the v1.1 research record and the added S19 proof, (ii) exact crossover-evaluation and feasibility-check code, (iii) a verification log and structured numerical data. No laboratory experiment was conducted for this revision.

References

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